Project Overview

AI Fraud Detector for Small Businesses · generated November 09, 2026

Small businesses are hit by fake orders, stolen cards and suspicious accounts, and rarely have a fraud team. This project builds an AI tool that automatically detects risky transactions so businesses don't lose money. It works like a smart fraud analyst that watches for unusual behaviour.

What the AI looks at

Signal family	Signals
Network activity	IP location, VPN / proxy usage, country mismatch, high-risk countries
Device behaviour	Emulator or virtual device, rooted device, timezone mismatch, device reused across many accounts
User behaviour	Too many / failed login attempts, rushed checkout, straight-to-checkout sessions, new accounts spending unusually large amounts
Transaction patterns	Strange or abnormal amounts, card-testing amounts, cards reused across accounts, odd hours

How the AI helps

- Learns from past normal and fraudulent behaviour
- Spots combinations of signals humans miss
- Gives each transaction a risk score (0-100) and an action: approve, review or block
- Shows **why** something looks risky, in plain language
- Helps businesses review and stop fraud early
- Ships with a QA layer that continuously tests accuracy, false positives, explanation quality and drift

Deliverables

- Data pipeline and synthetic dataset (2_Data)
- Trained model, risk scoring and explainability (3_Model)
- Evaluation & QA layer with prompt tests, golden fraud cases and model gates (3_Model/evaluation)
- REST API and database (4_Backend)
- Merchant dashboard and Streamlit QA console (5_Frontend)
- Architecture, performance, explainability reports and user manual (6_Documentation)